

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Ice-cold: {to prevent/slow down} any <u>enzyme</u> reactions (from the broken cell) (1) Isotonic: prevent lysis/prevent osmotic activity/or description of (1) NOT in context of cells			2	2		2
		(ii)		Increase reliability/minimise error due to chance/identify anomalous results/check repeatability/check consistency Reject accuracy/validity/fair test	1			1		1
		(iii)		Any three (x1) from 1. Tube 3 shows it was not the cold solution caused the DCPIP to decolourise(1) 2. Tube {3/4} shows that it was the chloroplasts that caused (the DCPIP to) decolourise (1) 3. Tube 4 shows that it is DCPIP that undergoes the colour change (1) 4. Both show DCPIP is not decolourised by light (1)		3		3		3
	(b)			Electrons are {released/excited/emitted} (1) By Photolysis splitting the water molecule/from photosystems/from reaction centre/from chlorophyll a (1) DCPIP is <u>reduced</u> (instead of NADP) (1)		2	1	3		
	(c)			Tube 2/tube in the dark (1) (Aerobic) respiration still occurs in the dark but not photosynthesis (1) But the DCPIP is not reduced/colourless/no colour change (1)			3	3		3
	(d)			The DCPIP is reduced due to the light dependent reaction/or description of (1) Carbon dioxide is only involved in the light independent reaction/Calvin cycle/or description of/carbon dioxide is not involved in the light dependent reaction(1)		2		2		
				Question 4 total	1	7	6	14	0	9